

# 20260320 Number Crunch

## Background and Problem Intuition

**Mathematical process:** Start from Bayes' rule, then convert inference into optimisation.

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})}, \quad (1)$$

$$p(\mathbf{y}) = \int p(\mathbf{y}, \boldsymbol{\theta}) d\boldsymbol{\theta}, \quad (2)$$

where  $p(\mathbf{y})$  is typically intractable.

Introduce a tractable variational density  $q(\boldsymbol{\theta})$  and write

$$\log p(\mathbf{y}) = \log p(\mathbf{y}) \int q(\boldsymbol{\theta}) d\boldsymbol{\theta} \quad (3)$$

$$= \int q(\boldsymbol{\theta}) \log p(\mathbf{y}) d\boldsymbol{\theta} \quad (4)$$

$$= \int q(\boldsymbol{\theta}) f(\boldsymbol{\theta}) d\boldsymbol{\theta}, \quad f(\boldsymbol{\theta}) \equiv \log p(\mathbf{y}) \quad (5)$$

$$= \mathbb{E}_q[f(\boldsymbol{\theta})] \quad (6)$$

$$= \mathbb{E}_q[\log p(\mathbf{y})] \quad (7)$$

$$= \mathbb{E}_q \left[ \log \frac{p(\mathbf{y}, \boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right] \quad (8)$$

$$= \mathbb{E}_q \left[ \log \left( \frac{p(\mathbf{y}, \boldsymbol{\theta})}{q(\boldsymbol{\theta})} \cdot \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right) \right] \quad (9)$$

$$= \mathbb{E}_q \left[ \log \frac{p(\mathbf{y}, \boldsymbol{\theta})}{q(\boldsymbol{\theta})} \right] + \mathbb{E}_q \left[ \log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right] \quad (10)$$

$$= \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})] + \mathbb{E}_q \left[ \log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right]. \quad (11)$$

**Sequence of events (plain-language):**

1. Start from  $\log p(\mathbf{y})$ , which is constant with respect to  $\boldsymbol{\theta}$ .
2. Multiply by 1 using normalisation of  $q$ :  $\int q(\boldsymbol{\theta}) d\boldsymbol{\theta} = 1$ .
3. Rewrite as an integral:  $\log p(\mathbf{y}) = \int q(\boldsymbol{\theta}) \log p(\mathbf{y}) d\boldsymbol{\theta}$ .
4. Recognise expectation form:  $\int q(\boldsymbol{\theta}) f(\boldsymbol{\theta}) d\boldsymbol{\theta} = \mathbb{E}_q[f(\boldsymbol{\theta})]$ .
5. Use Bayes identity inside the logarithm:  $p(\mathbf{y}) = \frac{p(\mathbf{y}, \boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})}$ .

6. Insert  $q(\boldsymbol{\theta})/q(\boldsymbol{\theta})$  to create ELBO and KL terms.
7. Split product and quotient with log rules, then split expectations linearly.
8. Identify

$$\text{ELBO}(q) = \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})]$$

and

$$\text{KL}(q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta} \mid \mathbf{y})) = \mathbb{E}_q \left[ \log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta} \mid \mathbf{y})} \right].$$

Define

$$\text{ELBO}(q) = \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\theta})] - \mathbb{E}_q[\log q(\boldsymbol{\theta})] \quad (12)$$

$$= \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\theta})] + H(q), \quad (13)$$

so that

$$\log p(\mathbf{y}) = \text{ELBO}(q) + \text{KL}(q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta} \mid \mathbf{y})). \quad (14)$$

Since  $\text{KL}(\cdot \parallel \cdot) \geq 0$ , maximising  $\text{ELBO}(q)$  is equivalent to minimising divergence to the true posterior.

## Key Notation

## Optimisation Problem Formulation

### Decision Variables

**Mathematical process:** Parameter counting and feasible-domain specification.

$$\boldsymbol{\phi} = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e), \quad (15)$$

$$\boldsymbol{\mu}_\beta \in \mathbb{R}^p, \quad \boldsymbol{\Sigma}_\beta \in \mathbb{S}_{++}^p, \quad a_e > 0, \quad b_e > 0, \quad (16)$$

$$d = p + \frac{p(p+1)}{2} + 2. \quad (17)$$

For  $p = 3$ :  $d = 3 + 6 + 2 = 11$ .

### Objective Function

**Mathematical process:** Expand  $\text{ELBO}(\boldsymbol{\phi})$  term-by-term under mean-field  $q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta}) q(\tau_e)$  with  $q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta)$  and  $q(\tau_e) = \text{Gamma}(a_e, b_e)$  (shape/rate parameterisation).

Write the five-term decomposition:

$$\text{ELBO}(\boldsymbol{\phi}) = T_1 + T_2 + T_3 + T_4 + T_5, \quad (18)$$

where

$$T_1 = \mathbb{E}_q[\log p(\mathbf{y} \mid \boldsymbol{\beta}, \tau_e)], \quad T_2 = \mathbb{E}_q[\log p(\boldsymbol{\beta})], \quad T_3 = \mathbb{E}_q[\log p(\tau_e)], \quad T_4 = -\mathbb{E}_q[\log q(\boldsymbol{\beta})], \quad T_5 = -\mathbb{E}_q[\log q(\tau_e)].$$

**Core building-block expectations** (used throughout):

$$\mathbb{E}_q[\tau_e] = \frac{a_e}{b_e}, \quad \mathbb{E}_q[\log \tau_e] = \psi(a_e) - \log b_e, \quad (19)$$

$$\mathbb{E}_q[\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2] = \|\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_\beta\|^2 + \text{tr}(\mathbf{X}^\top \mathbf{X} \boldsymbol{\Sigma}_\beta), \quad \mathbb{E}_q[\boldsymbol{\beta}^\top \boldsymbol{\beta}] = \boldsymbol{\mu}_\beta^\top \boldsymbol{\mu}_\beta + \text{tr}(\boldsymbol{\Sigma}_\beta). \quad (20)$$

The first two follow from Gamma moment formulae; the third uses  $\mathbb{E}_q[\boldsymbol{\beta}\boldsymbol{\beta}^\top] = \boldsymbol{\Sigma}_\beta + \boldsymbol{\mu}_\beta\boldsymbol{\mu}_\beta^\top$  and the trace-quadratic identity; the fourth is the same identity with  $\mathbf{A} = \mathbf{I}$ .

**$T_1$ : Expected log-likelihood**

From  $\mathbf{y} \mid \boldsymbol{\beta}, \tau_e \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \tau_e^{-1}\mathbf{I}_n)$ :

$$\log p(\mathbf{y} \mid \boldsymbol{\beta}, \tau_e) = \frac{n}{2} \log \tau_e - \frac{n}{2} \log(2\pi) - \frac{\tau_e}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2. \quad (21)$$

Taking the expectation and using mean-field independence ( $\tau_e \perp \boldsymbol{\beta}$  under  $q$ ):

$$T_1 = \frac{n}{2} (\psi(a_e) - \log b_e) - \frac{n}{2} \log(2\pi) - \frac{a_e}{2b_e} \left( \|\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_\beta\|^2 + \text{tr}(\mathbf{X}^\top \mathbf{X} \boldsymbol{\Sigma}_\beta) \right). \quad (22)$$

**$T_2$ : Expected log-prior on  $\boldsymbol{\beta}$**

From  $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda_\beta^{-1}\mathbf{I}_p)$ :

$$\log p(\boldsymbol{\beta}) = \frac{p}{2} \log \lambda_\beta - \frac{p}{2} \log(2\pi) - \frac{\lambda_\beta}{2} \boldsymbol{\beta}^\top \boldsymbol{\beta}, \quad (23)$$

$$T_2 = \frac{p}{2} \log \lambda_\beta - \frac{p}{2} \log(2\pi) - \frac{\lambda_\beta}{2} \left( \boldsymbol{\mu}_\beta^\top \boldsymbol{\mu}_\beta + \text{tr}(\boldsymbol{\Sigma}_\beta) \right). \quad (24)$$

**$T_3$ : Expected log-prior on  $\tau_e$**

From  $\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e)$  (shape/rate):

$$\log p(\tau_e) = \alpha_e \log \gamma_e - \log \Gamma(\alpha_e) + (\alpha_e - 1) \log \tau_e - \gamma_e \tau_e, \quad (25)$$

$$T_3 = \alpha_e \log \gamma_e - \log \Gamma(\alpha_e) + (\alpha_e - 1) (\psi(a_e) - \log b_e) - \gamma_e \frac{a_e}{b_e}. \quad (26)$$

**$T_4$ : Entropy of  $q(\boldsymbol{\beta})$**

The differential entropy of a  $p$ -variate Gaussian is

$$T_4 = H(q(\boldsymbol{\beta})) = \frac{p}{2} (1 + \log(2\pi)) + \frac{1}{2} \log |\boldsymbol{\Sigma}_\beta|. \quad (27)$$

With the Cholesky factor  $\boldsymbol{\Sigma}_\beta = \mathbf{L}\mathbf{L}^\top$  this becomes  $\log |\boldsymbol{\Sigma}_\beta| = 2 \sum_i \log L_{ii}$ , which is numerically stable.

**$T_5$ : Entropy of  $q(\tau_e)$**

The differential entropy of a  $\text{Gamma}(a, b)$  (shape/rate) distribution is

$$T_5 = H(q(\tau_e)) = a_e - \log b_e + \log \Gamma(a_e) + (1 - a_e) \psi(a_e). \quad (28)$$

**Full ELBO (up to constants in  $\phi$ )**

Dropping all terms that depend only on hyperparameters ( $\lambda_\beta, \alpha_e, \gamma_e$ ):

$$\text{ELBO}(\phi) = \frac{n}{2} (\psi(a_e) - \log b_e) - \frac{a_e}{2b_e} \left( \|\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_\beta\|^2 + \text{tr}(\mathbf{X}^\top \mathbf{X} \boldsymbol{\Sigma}_\beta) \right) \quad (T_1)$$

$$- \frac{\lambda_\beta}{2} (\boldsymbol{\mu}_\beta^\top \boldsymbol{\mu}_\beta + \text{tr}(\boldsymbol{\Sigma}_\beta)) \quad (T_2)$$

$$+ (\alpha_e - 1) (\psi(a_e) - \log b_e) - \gamma_e \frac{a_e}{b_e} \quad (T_3)$$

$$+ \frac{1}{2} \log |\boldsymbol{\Sigma}_\beta| \quad (T_4)$$

$$+ \log \Gamma(a_e) + (1 - a_e) \psi(a_e) - \log b_e + a_e + C, \quad (T_5)$$

where  $C$  absorbs all terms independent of  $\phi = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$ .

## Constraints

**Mathematical process:** Convert constrained variables to unconstrained parameters.

$$\Sigma_\beta = \mathbf{L}\mathbf{L}^\top, \quad \mathbf{L} \text{ lower triangular with } L_{ii} > 0, \quad (29)$$

$$a_e = e^{\eta_a}, \quad b_e = e^{\eta_b}, \quad \eta_a, \eta_b \in \mathbb{R}. \quad (30)$$

This guarantees  $\Sigma_\beta \succ 0$  and  $a_e, b_e > 0$  by construction.

## Computational Characteristics

**Mathematical process:** Leading-order complexity accounting.

$$\text{Gradient evaluation} : O(np^2), \quad (31)$$

$$\text{Newton step (Hessian build + solve)} : O(np^2 + d^3), \quad (32)$$

$$\text{BFGS update} : O(d^2). \quad (33)$$

## Test Case: Bayesian Linear Regression

**Mathematical process:** Specify generative model, variational family, and joint log-density components.

$$\mathbf{y} \mid \beta, \tau_e \sim \mathcal{N}(\mathbf{X}\beta, \tau_e^{-1}\mathbf{I}_n), \quad (34)$$

$$\beta \sim \mathcal{N}(\mathbf{0}, \lambda_\beta^{-1}\mathbf{I}_p), \quad (35)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e). \quad (36)$$

Useful joint log form (up to constants):

$$\log p(\mathbf{y}, \beta, \tau_e) = \frac{n}{2} \log \tau_e - \frac{\tau_e}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2 - \frac{\lambda_\beta}{2} \beta^\top \beta \quad (37)$$

$$+ (\alpha_e - 1) \log \tau_e - \gamma_e \tau_e + C. \quad (38)$$

## Overview of Optimisation Methods

### Methods considered

#### Baseline Method: CAVI<sup>1</sup>

**Mathematical process:** Coordinate-wise maximisation of ELBO.

$$\Sigma_\beta \leftarrow \left( \frac{a_e}{b_e} \mathbf{X}^\top \mathbf{X} \right)^{-1}, \quad (39)$$

$$\mu_\beta \leftarrow \frac{a_e}{b_e} \Sigma_\beta \mathbf{X}^\top \mathbf{y}, \quad (40)$$

$$a_e \leftarrow \alpha_e + \frac{n}{2}, \quad (41)$$

$$b_e \leftarrow \gamma_e + \frac{1}{2} \left( \|\mathbf{y} - \mathbf{X}\mu_\beta\|^2 + \text{tr}(\mathbf{X}^\top \mathbf{X} \Sigma_\beta) \right). \quad (42)$$

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<sup>1</sup>Blei, Kucukelbir, and McAuliffe (2017), Variational Inference: A Review for Statisticians (Journal of the American Statistical Association).

Each block update solves

$$q_j^{(t+1)} = \arg \max_{q_j} \text{ELBO}(q_j, q_{-j}^{(t)}), \quad (43)$$

so ELBO is non-decreasing across iterations.

## Mapping ENEL445 Methods to VI Problem

**Mathematical process:** Express optimisation updates on  $\phi$ .

**Gradient ascent**

$$\phi^{(k+1)} = \phi^{(k)} + \alpha_k \nabla \text{ELBO}(\phi^{(k)}). \quad (44)$$

**Newton's method**

$$\phi^{(k+1)} = \phi^{(k)} + \alpha_k [\nabla^2 \text{ELBO}(\phi^{(k)})]^{-1} \nabla \text{ELBO}(\phi^{(k)}). \quad (45)$$

**BFGS (inverse form)**

$$\mathbf{s}_k = \phi_{k+1} - \phi_k, \quad \mathbf{y}_k = \nabla \text{ELBO}(\phi_{k+1}) - \nabla \text{ELBO}(\phi_k), \quad (46)$$

$$\rho_k = (\mathbf{y}_k^\top \mathbf{s}_k)^{-1}, \quad (47)$$

$$\mathbf{B}_{k+1} = (\mathbf{I} - \rho_k \mathbf{s}_k \mathbf{y}_k^\top) \mathbf{B}_k (\mathbf{I} - \rho_k \mathbf{y}_k \mathbf{s}_k^\top) + \rho_k \mathbf{s}_k \mathbf{s}_k^\top. \quad (48)$$

## Solution Plan

### Phase 1: Data Generation and Baseline

**Mathematical process:** Derive and verify all analytical quantities required before coding.

$$\text{Deliverables : } \text{ELBO}(\phi), \nabla \text{ELBO}(\phi), \nabla^2 \text{ELBO}(\phi), \quad (49)$$

$$\text{CAVI updates, and numerical checks against finite differences.} \quad (50)$$

### Phase 2: Implementation (in progress)

#### Success Metrics

**Mathematical process:** Define quantitative stopping and agreement criteria.

$$\text{Function tolerance: } |\text{ELBO}^{(k+1)} - \text{ELBO}^{(k)}| < 10^{-6}, \quad (51)$$

$$\text{Gradient tolerance: } \|\nabla \text{ELBO}(\phi^{(k)})\|_2 < 10^{-6}, \quad (52)$$

$$\text{Method agreement: } |\text{ELBO}_{\text{method}}^* - \text{ELBO}_{\text{CAVI}}^*| \leq \varepsilon. \quad (53)$$

## Progress To Date

### Work in progress using an iterative / Agile method

**Mathematical process identified as drafted/partial:**

$$\text{(i) ELBO derivation from model assumptions,} \quad (54)$$

$$\text{(ii) CAVI closed-form updates,} \quad (55)$$

$$\text{(iii) analytical gradients for } d = 11, \quad (56)$$

$$\text{(iv) Hessian block structure,} \quad (57)$$

$$\text{(v) preliminary BFGS and penalty formulations.} \quad (58)$$

## Preliminary Results

**Mathematical process:** Local structure checks before implementation.

$$\nabla_{\boldsymbol{\mu}_\beta} \text{ELBO} \propto \boldsymbol{X}^\top (\boldsymbol{y} - \boldsymbol{X} \boldsymbol{\mu}_\beta), \quad (59)$$

$$\text{Hessian block sign checks} \Rightarrow \text{local concavity in selected directions}, \quad (60)$$

$$d = 11 \Rightarrow \text{Newton linear-algebra cost is tractable at this scale.} \quad (61)$$

## Next Steps

## Alignment with Course Requirements

## Timeline

## Summary